

Jason Jingzhou Liu

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EDUCATION

- **Carnegie Mellon University** Aug 2024 - Present
Ph.D in Robotics Pittsburgh, USA
 - Advised by Prof. Deepak Pathak and Prof. Ruslan Salakhutdinov (GPA: 4.0/4.0)
- **University of Toronto** Sep 2019 - Apr 2024
Bachelor of Applied Science in Engineering Science, Robotics Major Toronto, Canada
 - Rank 1/250 (GPA: 3.98/4.0)

PUBLICATIONS

- [1] S. Oh*, **J.J. Liu***, T. Tao*, P. Han, K. Shaw, Satoshi Funabashi, Ruslan Salakhutdinov, D. Pathak. "FACTR 2: Learning External Force Sensing for Commodity Robot Arms Improves Policy Learning". *In Submission*, 2026.
- [2] **J.J. Liu***, J. Yang*, R. Liu, A. Wang, K. Shaw, D. Pathak, R. Salakhutdinov. "COWBOY A Scalable Sim-to-Real Framework for Learning Contextual Whole-Body Manipulation". *In Submission*, 2026.
- [3] J. Yang*, **J.J. Liu***, Y. Li, Y. Khaky, K. Shaw, D. Pathak. "Deep Reactive Policy: Learning Reactive Manipulator Motion Planning for Dynamic Environments". *Conference on Robot Learning (CoRL)*, 2025.
- [4] **J.J. Liu***, Y. Li*, K. Shaw, T. Tao, R. Salakhutdinov, D. Pathak. "FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning". *Robotics: Science and Systems (RSS)*, 2025.
- [5] T. Tao*, M. Srirama*, **J.J. Liu**, K. Shaw, D. Pathak. "DexWild: Dexterous Human Interactions for In-the-Wild Robot Policies". *Robotics: Science and Systems (RSS)*, 2025.
- [6] R. Singh, **J.J. Liu**, K. V. Wyk, Y. Chao, J. Lafleche, F. Shkurti, N. Ratliff, A. Handa. "Synthetica: Large Scale Synthetic Data Generation for Robot Perception". *International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [7] Q. Yu*, M. Moghani*, K. Dharmarajan, V. Schorp, W. Panitch, **J.J. Liu**, K. Hari, H. Huang, M. Mittal, K. Goldberg, A. Garg. "ORBIT-Surgical: An Open-Simulation Framework for Accelerated Learning Environments in Surgical Autonomy". *International Conference on Robotics and Automation (ICRA)*, 2024.
- [8] S. Zhang*, Y. Qiao*, G. Zhu*, E. Heiden, D. Turpin, **J.J. Liu**, M. C. Lin, M. Macklin, A. Garg. "HandyPriors: Physically Consistent Perception of Hand-Object Interactions with Differentiable Priors". *International Conference on Robotics and Automation (ICRA)*, 2024.
- [9] A. Handa*, A. Allshire*, V. Makoviychuk*, A. Petrenko*, R. Singh*, **J.J. Liu***, D. Makoviichuk, K. V. Wyk, A. Zhurkevich, B. Sundaralingam, Y. Narang, J. Lafleche, D. Fox, G. State. DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality. *International Conference on Robotics and Automation (ICRA)*, 2023.
- [10] D. Turpin, T. Zhong, S. Zhang, G. Zhu, **J.J. Liu**, R. Singh, E. Heiden, M. Macklin, S. Tsogkas, S. Dickinson, A. Garg. "Fast-Grasp'D: Dexterous Multi-finger Grasp Generation Through Differentiable Simulation". *International Conference on Robotics and Automation (ICRA)*, 2023.
- [11] M. Attarian, M. Asif, **J.J. Liu**, R. Hari, A. Garg, I. Gilitschenski, J. Thompson. "Geometry Matching for Multi-Embodiment Grasping". *Conference on Robot Learning (CoRL)*, 2023.
- [12] M. Mittal, C. Yu, Q. Yu, **J.J. Liu**, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, A. Garg. "Orbit: A unified simulation framework for interactive robot learning environments". *Robotics and Automation Letters (RA-L)*, 2023.

EXPERIENCE

- **NVIDIA** May 2025 - Aug 2025
Research Scientist Intern
 - Worked on reinforcement learning for fine-tuning BC policies for dexterous, long-horizon manipulation.
 - Advised by Nathan Ratliff, Ankur Handa, and Karl Van Wyk.
- **Robot Vision and Learning Lab, University of Toronto** Sep 2023 - Apr 2024
Undergraduate Researcher with Prof. Florian Shkurti
 - Completed [undergraduate thesis](#) on large-scale RL for manipulator collision-free motion planning.
 - Achieved SOTA learning-based motion planning results via training RL in Isaac Gym with Geometric Fabrics in procedurally generated obstacle environments with novel scene conditioning method ([video](#)).
- **NVIDIA** Jan 2022 - Aug 2023
Robotics Simulation and Deep Learning Intern (Isaac Gym Team)

- **In-Hand Manipulation:** Co-first authored [DeXtreme](#), demonstrating human-level dexterity on anthropomorphic robot hands with large-scale RL and sim-to-real transfer. Implemented a pipeline to generate 20 mil + simulated images and trained Mask-RCNN object pose estimation for the RL policy.
 - **Object Detection:** Co-authored [Synthetica](#), achieving SOTA object-detection via training on large-scale simulation images, accepted to IROS 2025. Led the synthetic image generation pipeline (20 mil+ images). Achieved [first-place](#) on the BOP object-detection leader board.
 - **RL Training:** Co-led [OmniverseIsaacGymEnvs](#), a collection of 10+ RL robotic tasks in simulation (manipulation, locomotion tasks, etc.). The codebase has 990+ stars and 231+ forks on GitHub.
 - **Simulation Development:** Core developer of Isaac Sim. Main contributions include 30× the max number of vectorized environments in simulation (from 500 to 16000), increased RL training speed by 10×, leading the development of physics-based domain randomization APIs for RL. I also led the development of [URDF Importer](#) and [MJCF Importer](#) for loading simulation assets in Isaac Sim.
- **Vector Institute** Jan 2022 - Aug 2023
Robotics Researcher with Prof. Animesh Garg
- **Robotics Simulation:** Co-authored and core developer of [Orbit](#) (later renamed to [Isaac Lab](#)), a unified simulation framework for robot learning, accepted to RA-L. Implemented IK, OSC, RMPFlow, and various RL environments in simulation. Performed sim-to-real experiments with a robot arm-hand setup.
 - **Dexterous Teleoperation:** Led the development of one of the first vision-based robotic hand-arm teleoperation system ([video](#)). Pipeline includes a robust, real-time, RGB-based hand-pose estimator, kinematic retargeting optimization, and reactive task-space motion controllers on the real robot.
 - **Dexterous Grasp Generation:** Co-authored [Fast-Grasp'D](#), a grasp generator for multi-fingered robotic hands via differentiable simulation, accepted to ICRA 2023. Performed sim-to-real experiments using a hand-arm system to perform grasping.

HONORS AND AWARDS

- **Best Paper Award** Oct 2025
LeaPRiDE Workshop, IROS 2025
 - Awarded for "Deep Reactive Policy: Learning Reactive Manipulator Motion Planning for Dynamic Environments".
- **Best Paper Award** Jun 2025
EgoAct Workshop, RSS 2025
 - Awarded for "DexWild: Dexterous Human Interactions for In-the-Wild Robot Policies".
- **W.S. Wilson Medal** May 2024
University of Toronto
 - Awarded for graduating first in class at the University of Toronto Engineering.
- **Best Undergraduate Thesis** Apr 2024
University of Toronto
 - Awarded to a final year Engineering Science student whose thesis demonstrates exceptional innovation.
- **Engineering Science Award of Excellence** Apr 2024
University of Toronto
 - Awarded for achieving a cGPA of 3.9/4.0 or higher in the Engineering Science program.
- **Charles Edwin Trim Scholarship** Aug 2023
University of Toronto
 - Awarded based on academic excellence.
- **Dharma Master Chuk Mor Memorial Scholarship** Aug 2022
University of Toronto
 - Awarded based on academic excellence.
- **Sullivan Memorial Scholarship** Aug 2021
University of Toronto
 - Awarded based on academic excellence.
- **Shaw Scholarship** Aug 2020
University of Toronto
 - Awarded to one student of Engineering Science who demonstrates high academic achievement, leadership skills, and design capabilities.
- **C. David Naylor University Scholarship** Aug 2019
University of Toronto
 - Awarded to 10 students based on academic merit and demonstrated leadership excellence.

TECHNICAL SKILLS

- **Robotics:** Simulation (Isaac Sim/Gym, Pybullet, etc.), Sim-to-Real, Advanced Controls (Geometric Fabrics, RMP, OSC, Whole Body Control, etc.), Motion Planning, State Estimation (Kalman Filter, SLAM, etc.)
- **Machine Learning:** RL (PPO, etc.), Behavior Cloning (ACT, VLA, etc.), Transformers, Flow Matching, etc.
- **Software:** C++, C, Python (PyTorch, JAX, ROS, etc.),
- **Mechanical:** SOLIDWORKS (CSWP), 3D Printing